

Jonnathan Singh Alvarado, Ph.D.

Contact

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Education

Postdoctoral Fellow, Neurobiology, Harvard University 2021-Current

Ph.D., Neurobiology, Duke University 2014-2021

B.S., Microbiology, University of Florida 2009-2013

Research Experience

Postdoctoral Fellow 2021-Current

Dr. Mark Andermann, Beth Israel Deaconess Medical Center

Graduate Student, Duke University 2014-2021

Dr. Richard Mooney, Department of Neurobiology

Research Assistant, Duke University 2013-2014

Dr. Cagla Eroglu, Department of Cell Biology

Undergraduate Researcher, University of Florida 2012-2013

Dr. Stephanie Karst, Department of Virology

Undergraduate Researcher, University of Florida 2011-2012

Dr. Brent Reynolds, Department of Cancer Biology

Publications

EASI-PASS: an accessible pipeline for linking functional imaging and mRNA profiling. **Singh Alvarado J**, Massengill C, Stern J, Amsalem O, Ventura B, Jang A, Cook S, Veliche A, Sunkavalli P, Patel D, Colaccino J, Evans K, Wang Y, Andermann ML. *bioRxiv*. 2026.

The role of modern neuroscience in the renaissance of psychedelic research. Evans KE, Massengill C, **Singh Alvarado J**. *Revista Costarricense de Psicología*. 2025.

A sensitive soma-localized red fluorescent calcium indicator for in vivo imaging of neuronal populations at single-cell resolution. Zhou S, Zhu Q, Eom M, Fang S, Subach O, Ran C, **Singh Alvarado J**, Sunkavalli P, Yuanping Dong, Wang Y, Wang Z, Sun X, Fedor V Subach, Kiryl D Piatkevich. *PLoS Biology*. 2025.

Transient cAMP production drives rapid and sustained spiking in brainstem parabrachial neurons to suppress feeding. **Singh Alvarado J***, Lutas A*, Madara JC, Isaac J, Lommer C, Massengill C, Andermann ML. *Neuron*. 2024.

Divergent projections from Locus Coeruleus to the cortico-basal ganglia system and ventral tegmental area of the zebra finch. **Singh Alvarado J***, Hatfield J*, Mooney R. *Journal of Comparative Neurology*. 2022.

Neural dynamics underlying birdsong practice and performance. Singh Alvarado J, Goffinet J, Michael V, Liberti W, Hatfield J, Gardner T, Pearson J, & Mooney R. *Nature*. 2021.

Building Opportunities and Overtures in Science and Technology: Establishing an Early Intervention, Multi-level, Continuous STEM Pathway Program. Garcia J, Gibson R, Breslav A, Singh Alvarado J, Seas A, Maduka A, et al. *Journal of STEM Outreach*. 2021.

MIN1PIPE: A Miniscope 1-Photon-Based Calcium Imaging Signal Extraction Pipeline. Lu J, Li C, Singh Alvarado J, Zhe Charles Zhou, Flavio Fröhlich, Mooney R, Wang F. *Cell Rep*. 2018.

Expressing mutant huntingtin in the songbird basal ganglia disrupts cortico-basal ganglia networks and vocal sequences. Tanaka M, Singh Alvarado J, Murugan M, and Mooney R. *Proc. Natl. Acad. Sci. USA*. 2016.

Astrocytes refine cortical connectivity at dendritic spines. Risher WC, Patel S, Kim IH, Uezu A, Pilaz LJ, Singh Alvarado J, Calhan OY, Silver DL, Stevens B, Soderling SH, and Eroglu C. *eLife*. 2014.

Rapid Golgi Analysis Method for Efficient and Unbiased Classification of Dendritic Spines. Risher WC, Ustunkaya T, Singh Alvarado J, Eroglu C. *PLoS ONE*. 2014.

* denotes equal contribution.

Talks

Outstanding Postdoc Talk – HMS Neurobiology Departmental Retreat	2026
Optogenetic Approaches to Understanding Neural Circuits and Behavior – Lucca, Italy	2026
Systems Neuroscience Club – Harvard Medical School	2026
Departmental Pizza Talks – Harvard Medical School Department of Neurobiology	2025
Department Lab Results Seminar – Boston Children’s Hospital Kirby Neurobiology Center	2024
Endocrinology Division Meeting – BIDMC	2024
Special Seminar – University of Pittsburgh & Carnegie Mellon University	2024
SFN Minisymposium – Peripheral mechanisms of energy & metabolism	2023
Special Seminar – Champalimaud Centre for the Unknown	2023
Endocrinology Division Meeting – BIDMC	2022
Joint Lab Meeting – Carnegie Mellon University Systems Club	2022
COSYNE (6% of abstracts selected)	2021
Special Seminar – Columbia University	2021
Virtual Dopamine Conference (ViDA)	2021
University of Washington and Allen Institute Neuromodulation Club	2020
Neuromodulation Gordon Research Conference	2019
Duke University Brown Bag Seminars	2018
Duke Institute for Brain Sciences Symposium	2017
Duke TEDx	2016

Fellowships and Awards

BBRF Young Investigator Grant, Brain & Behavior Research Foundation	2026-2028
Harvard Neurobiology David Hubel Outstanding Postdoc Award	2026
The Transmitter “Rising Stars of Neuroscience”	2025
NINDS K99/R00 (parent mechanism), deferred	2025
MOSAIC NINDS K99/R00 Award	2024
Harvard Brain Initiative Postdoctoral Pioneer Award	2024-2025
Harvard Mind, Brain, and Behavior (MBB) postdoctoral fellowship	2022-2024
Graduate student training enhancement grant (GSTEG) awardee	2019
Simons Foundation Neuromodulation GRC – Outstanding Talk Award	2019
Cold Spring Harbor Laboratories – Neural data science course	2019
Duke Institute for Brain Sciences Symposium – Best Talk	2017
Ruth K. Broad Research Fellowship	2016
Biosciences Collaborative for Research Engagement (BioCoRe) Scholar	2015
Latin American & Caribbean Scholarship (4-year scholarship)	2009-2013

Leadership

Gordon Research Seminar Co-chair 2025-Present
Organizer for the 2027 Neuromodulation Gordon Research Seminar, Barcelona.

HPREP mentor 2024-Present
Lecturer and hands-on project coordinator for STEM outreach program HPREP at Harvard.

NeuroLaunchpad 2021-2022
Created and received funding for a trainee-centered nationwide seminar series to give exposure to burgeoning scientists, resulting in over 1000 unique viewers.

BOOST Coach 2014-2021
Developed year-long STEM-based research programs and hands-on learning curriculums for low-income middle and high school students.

Committee for Reimagining Doctoral Education at Duke 2018-2020
One of three Duke graduate students selected to serve. Met monthly with 30 faculty, the provost, and the research training Dean to construct a strategic plan for restructuring doctoral education.

Other Service 2014-2021
Neurobiology Diversity and Inclusion taskforce, Neurobiology Steering Committee, Deans Mentoring and Gordon G Hammes mentoring award committees, Faculty advancement committee, Alfred P Sloan Scholars committee.

Rex Scholar 2016-2017
Assisted in developing software and virtual lectures to be widely distributed in NC public schools.

TECHO – Costa Rica NGO 2007-2011
Directed teams of volunteers in the construction of emergency transitional homes for families in need.